



UB-7427

**Integrated M. C. A. (Sem. II)
Examination, June - 2025**

IMCA-202T

Computer Architecture

Time Allowed : Three Hours

Maximum Marks : 70

Min. Marks : 28

***Note : This question paper contains two
sections as under :***

Section-A**Max. Marks - 20**

This section will have 10 questions, having 02 questions from each unit with no internal choice. Answer in about 20 words. All questions will have equal marks. (10×02=20)

Section-B**Max. Marks-50**

This section will have 10 questions, having 02 questions from each unit, one question to be attempted from each unit. Words limit approximately 400-500 words. Questions may have sub divisions. All questions will have equal marks. (10×05=50)

SECTION - A

- 1 (i) Explain instruction set.
- (ii) Write down Flynn's classification of computer.
- (iii) What is a bus? Explain.
- (iv) What is accumulator (AC)? What is its use?
- (v) What is the use of Program Counter (PC)?
- (vi) What are the function of Control Unit (CU)?
- (vii) What is Cache?
- (viii) Differentiate b/w Machine and Assembly language.
- (ix) What are external devices? Give example.
- (x) What is cache hit and cache miss?

SECTION - B

UNIT - I

- 2** What is Von-neumann Architecture? Explain how a program is executed in this architecture.

OR

Explain the instruction cycle with the help of a neat diagram. Differentiate between fetch and execute cycle.

UNIT - II

- 3** Describe different types of addressing modes with example. Why are addressing modes important in instruction design?

OR

Explain RISC and CISC architecture. Compare their features and performance.

UNIT - III

- 4** What is associative memory? Describe its working with a suitable diagram.

OR

Explain cache memory organization. How does cache hit/miss and hit ratio affect system performance?

UNIT - IV

- 5** Compare and contrast programmed I/O, Interrupt - Driven I/O, and Direct Memory Access (DMA).

OR

Explain the structure and functions of I/O modules with examples of external devices.

UNIT - V

- 6** Explain the construction and working of CMOS design. Why is CMOS preferred in modern IC design?

OR

Differentiate between current - controlled and voltage - controlled switches. Explain the role of Bipolar Transistor and MOSFET.
